

Gyuseong Lee

RESEARCH SCIENTIST, LG ELECTRONICS

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Summary

Computer vision and generative AI researcher with industry experience at LG Electronics. Co-authored papers published in CVPR, ICCV (100+ citations), ICLR (oral), ECCV, and WACV, with total citations exceeding 300. Recognized as Outstanding Reviewer at ECCV 2024. Research focuses on synthetic data utilization, controllable generation of diffusion models, robust fine-tuning of large models, and visual correspondence.

Experience

LG Electronics

Seoul, S.Korea

COMPUTER VISION RESEARCH SCIENTIST

Feb. 2024 -

- Built visual synthetic data generation pipelines (diffusion image generation + filtering) to improve classification accuracy by 90-95% on internal models.
- Fine-tuned large multimodal models for small image recognition (improving accuracy from 66% to 70%) and for referring object detection (reducing inference latency) using synthetic data.
- Contributed to the development of open-vocabulary detection models, matching our internal model's performance with that of state-of-the-art models.

Educations

Korea University

Seoul, S.Korea

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Mar. 2022 - Feb. 2024

- Advisor: Prof. Seungryong Kim

Korea University

Seoul, S.Korea

B.S. IN ELECTRICAL ENGINEERING

Mar. 2016 - Feb. 2022

Research Interests

- Diffusion-based Generative Models: Guidance Methods, Controllability, and Data Generation
- Robust Fine-tuning of Large Models
- Semi- & Self-Supervised Learning
- Visual Correspondence

Publications

Diffusion Model for Dense Matching

ICLR 2024 Oral Presentation

JISU NAM, **GYUSEONG LEE**, SUNWOO KIM, HYEONSU KIM, HYOUNGWON CHO, SEYEON KIM, SEUNGRYONG KIM

- **Topics:** Diffusion Generative Models, Image Matching

Improving Sample Quality of Diffusion Models Using Self-Attention Guidance

ICCV 2023

SUSUNG HONG, **GYUSEONG LEE**, WOOSOOK JANG, SEUNGRYONG KIM

- **Topics:** Diffusion Generative Models, Classifier-Free Guidance

Domain Generalization Using Large Pretrained Models With Mixture-of-Adapters

WACV 2025

GYUSEONG LEE*, WOOSOOK JANG*, JIN HYEON KIM, JAEWOO JUNG, SEUNGRYONG KIM

- **Topics:** Domain Generalization, Robust Fine-tuning

Depth-Aware Guidance with Self-Estimated Depth Representations of Diffusion Models

Pattern Recognition

GYEONGNYEON KIM*, WOOSOOK JANG*, **GYUSEONG LEE***, SUSUNG HONG, JUNYOUNG SEO, SEUNGRYONG KIM

- **Topics:** Diffusion Generative Models, Sampling Guidance, Diffusion Features

* Equal contribution.

MIDMs: Matching Interleaved Diffusion Models for Exemplar-based Image Translation

AAAI 2023

JUNYOUNG SEO*, **GYUSEONG LEE***, SEOKJU CHO, JIYOUNG LEE, SEUNGRYONG KIM (CO-FIRST AUTHOR)

- **Topics:** Diffusion Generative Models, Image-to-Image Translation, Image Matching

Towards Flexible Inductive Bias via Progressive Reparameterization Scheduling

ECCV 2022 VIPriors Workshop

YUNSUNG LEE*, **GYUSEONG LEE***, KWANGROK RYOO*, HYOUNG GO*, JIHYE PARK*, SEUNGRYONG KIM

- **Topics:** Model Architecture, Vision Transformer

ConMatch: Semi-supervised Learning with Confidence-Guided Consistency Regularization

ECCV 2022

JIWON KIM, YOUNGJO MIN, DAEHWAN KIM, **GYUSEONG LEE**, JUNYOUNG SEO, KWANGROK RYOO, SEUNGRYONG KIM

- **Topics:** Image Classification, Semi-Supervised Learning

Semi-Supervised Learning of Semantic Correspondence with Pseudo-Labels

CVPR 2022

JIWON KIM*, KWANGROK RYOO*, JUNYOUNG SEO*, **GYUSEONG LEE***, DAEHWAN KIM, HANSANG CHO, SEUNGRYONG KIM (CO-FIRST AUTHOR)

- **Topics:** Image Matching, Semi-Supervised Learning

AggMatch: Aggregating Pseudo Labels for Semi-Supervised Learning

arXiv preprint

JIWON KIM, KWANGROK RYOO, **GYUSEONG LEE**, SEOKJU CHO, JUNYOUNG SEO, DAEHWAN KIM, HANSANG CHO, SEUNGRYONG KIM

Jan. 2022

- **Topics:** Image Classification, Semi-Supervised Learning

Awards & Scholarships

2024 **Outstanding Reviewer**, ECCV 2024

Milano, Italy

2023 **Best Paper Award**, 33th Artificial Intelligence and Signal Processing, IEIE

Seoul, S.Korea

2022 **Best Paper Award**, 32nd Joint Conference of Signal Processing, IEIE

2020 **Dean's List**, Korea University

2021 **Semester High Honors**, Korea University

National Science & Technology Scholarship, Korea Student Aid Foundation

2016-2017 **Semester High Honors**, Korea University

Skills

Core Python PyTorch Docker Git LaTeX Gerrit

Previously used C MATLAB Verilog

Projects

On-device Scene Understanding & Gesture Interaction for Pro-active Home Agents

LG Electronics (PoC)

- Prototyped a Raspberry Pi 5 home agent featuring motion detection, image captioning, and LLM-based scene-state recognition; defined three agent states (lying, sitting, fallen) and implemented real-time hand-gesture control for state transitions.

Feb. 2025 – Mar. 2025

Multi-domain Generalizable Anomaly Classification (Industry Project)

Samsung Electro-Mechanics

- Investigated OOD robustness and model bias; developed domain generalization and label-efficient adaptation leveraging large pretrained models for anomaly classification.

Sep. 2022 – Apr. 2023

Research Infrastructure & GPU Monitoring System

Korea University

- Developed a centralized dashboard to monitor health and resource utilization across the lab's multi-GPU servers using DCGM-Exporter, Prometheus/Grafana, and Uptime-Kuma to address hardware issues.

Sep. 2022 – Dec. 2023

* Equal contribution.

Semi-supervised & continual learning for anomaly classification (Industry Project)

Samsung Electro-Mechanics

- Utilized a contrastive-based loss function to improve the classification accuracy of fine-grained, long-tailed defective samples in a semi-supervised setting.
- Applied knowledge distillation to preserve model performance over time in continual learning scenarios.

July. 2021 - May. 2022

Peer Review

- Selected as an **outstanding reviewer at ECCV 2024**
- Served as a reviewer for top-tier conferences including CVPR, ICCV/ECCV, AAAI, and WACV (2023-)